# BENJAMIN ATTAL

benattal.github.io • battal@andrew.cmu.edu • 347-601-8715

## EDUCATION

**Carnegie Mellon Robotics Institute** *September 2019 - June 2025 (Expected)*
*Ph.D. in Robotics* GPA: 4.0

**Brown University** *September 2014 - May 2019*
*B.S. in Applied Math and Computer Science, M.S. in Applied Math* GPA (within major): 3.9

## RELEVANT WORK EXPERIENCE

**Google Research** *Summer 2023*
*PhD Student Researcher with Pratul Srinivasan*

**Facebook Computational Photography** *Summer 2021, 2022*
*PhD Student Researcher with Changil Kim*

**CMU Light Transport Lab** *Fall 2019 - Present*
*PhD Student with Matt O'Toole*

**Brown Visual Computing Lab** *Fall 2018 - Fall 2019*
*Graduate Student Researcher with James Tompkin*

**Light** *Fall 2018 - Spring 2019*
*Research Intern*

## AWARDS

**Uber Fellowship** *2021*

**Meta Fellowship** *2023-2025*
AR/VR Computer Graphics

## SELECTED PUBLICATIONS

***Neural Inverse Rendering from Propagating Light.*** **CVPR 2025 (Oral)**
*Anagh Malik*, Benjamin Attal*, Andrew Xie, Matt O'Toole, David Lindell.*

***Flash Cache: Reducing Bias in Radiance Cache Based Inverse Rendering.*** **ECCV 2024 (Oral)**
*Benjamin Attal, Dor Verbin, Ben Mildenhall, Peter Hedman, Jon Barron, Matt O'Toole, Pratul Srinivasan.*

***Flowed Time of Flight Radiance Fields.*** **ECCV 2024**
*Mikhail Okunev*, Marc Mapeke*, Benjamin Attal, Christian Richardt, Matt O'Toole, James Tompkin.*

***Neural Fields for Structured Lighting.*** **ICCV 2023**
*Aarushi Shandilya, Benjamin Attal, Christian Richardt, James Tompkin, Matt O'Toole.*

***HyperReel: High-Fidelity 6-DoF Video with Ray-Conditioned Sampling.*** **CVPR 2023 (Highlight)**
*Benjamin Attal, Jia-Bin Huang, Christian Richardt, Michael Zollhöfer, Johannes Kopf, Matthew O'Toole, Changil Kim.*

***Learning Neural Light Fields with Ray-Space Embedding Networks.*** **CVPR 2022**
*Benjamin Attal, Jia-Bin Huang, Michael Zollhöfer, Johannes Kopf, Changil Kim.*

***Towards Mixed-State Coded Diffraction Imaging.*** **TPAMI 2022**
*Benjamin Attal, Matt O'Toole.*

***TöRF: Time-of-Flight Radiance Fields for Dynamic Scene View Synthesis.*** **NeurIPS 2021**
*Benjamin Attal, Eliot Laidlaw, Aaron Gokaslan, Changil Kim, Christian Richardt, James Tompkin, Matt O'Toole.*

***MatryODShka: Real-time 6DoF Video View Synthesis using Multi-Sphere Images.*** **ECCV 2020 (Oral)**
*Benjamin Attal, Selena Ling, Aaron Gokaslan, Christian Richardt, James Tompkin.*

## TALKS

**ECCV 2024** *Summer 2024*
*Flash Cache: Reducing Bias in Radiance Cache Based Inverse Rendering*

**ICCP 2022** *Summer 2022*
*Towards Mixed-State Coded Diffraction Imaging*

**Google (Invited)** *Spring 2022*
*Learning Neural Light Fields with Ray-Space Embedding Networks*

**ECCV 2020** *Summer 2020*
*Real-time 6DoF Video View Synthesis using Multi-Sphere Images*

## SERVICE

**Reviewer**

- SIGGRAPH
- CVPR
- NeurIPS
- ICCV, ECCV
- ACM Transactions on Graphics
- Computer Graphics Forum

## TEACHING

**Teaching Assistant**

- Computer Vision (CMU 16385)
- Learning for 3D Vision (CMU 16889)
- Computer Graphics (Brown CSCI 1230)
- 2D Game Engine Development (Brown CSCI 1950N)

**Head Teaching Assistant**

- 3D Game Engine Development (Brown CSCI 1950U)